

SESSION 2 - LAB EXPERIMENTS

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1. Find Unique Elements

```
#include <iostream>

#include <vector>
#include <unordered_set>
using namespace std;

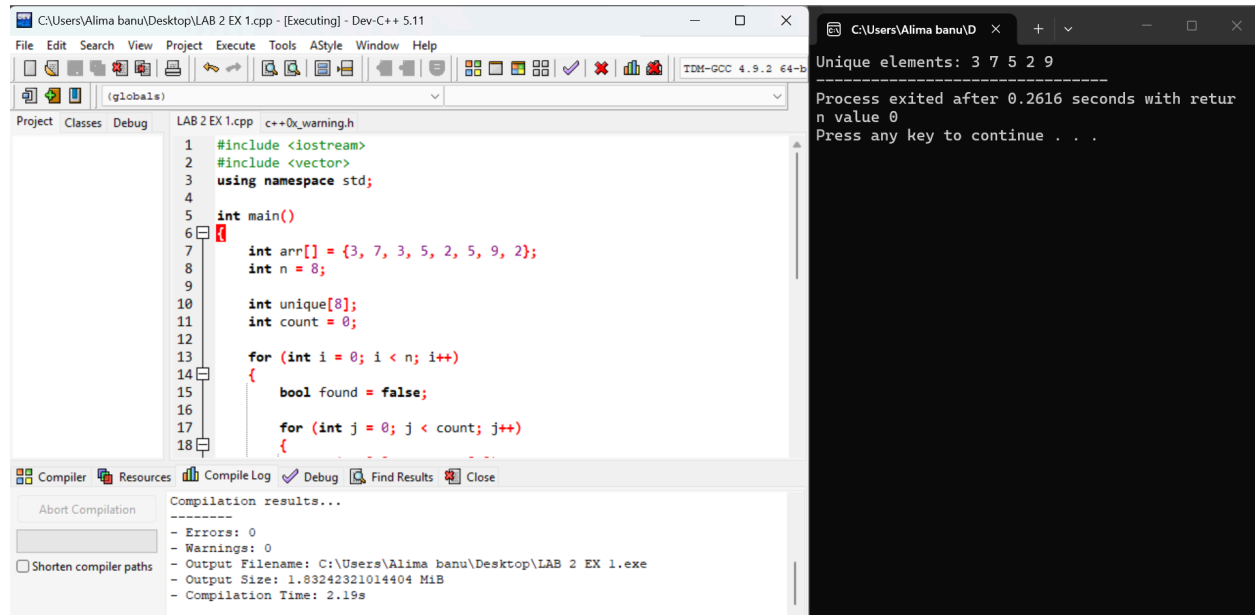
int main() {
    vector<int> arr = {3, 7, 3, 5, 2, 5, 9, 2};

    unordered_set<int> seen;
    vector<int> unique;

    for (int x : arr) {
        if (seen.find(x) == seen.end()) {
            seen.insert(x);
            unique.push_back(x);
        }
    }
}
```

```
    }  
}  
  
cout << "Unique elements: ";  
  
for (int x : unique)  
    cout << x << " ";  
  
return 0;  
}
```

OUTPUT :



The screenshot shows a C++ IDE with a source code editor on the left and a console window on the right. The source code in the editor is as follows:

```
1 #include <iostream>
2 #include <vector>
3 using namespace std;
4
5 int main()
6 {
7     int arr[] = {3, 7, 3, 5, 2, 5, 9, 2};
8     int n = 8;
9
10    int unique[8];
11    int count = 0;
12
13    for (int i = 0; i < n; i++)
14    {
15        bool found = false;
16
17        for (int j = 0; j < count; j++)
18        {
```

The console window on the right displays the output of the program:

```
Unique elements: 3 7 5 2 9
-----
Process exited after 0.2616 seconds with return value 0
Press any key to continue . . .
```

Below the source code editor, the 'Compiler' tab is active, showing the following compilation results:

```
Compilation results...
-----
- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\Alima banu\Desktop\LAB 2 EX 1.exe
- Output Size: 1.83242321014404 MiB
- Compilation Time: 2.19s
```

Ex 2. Bubble Sort

```
#include <iostream>
using namespace std;
```

```
int main()
{
    int arr[] = {5, 2, 8, 1, 3};
    int n = 5;
```

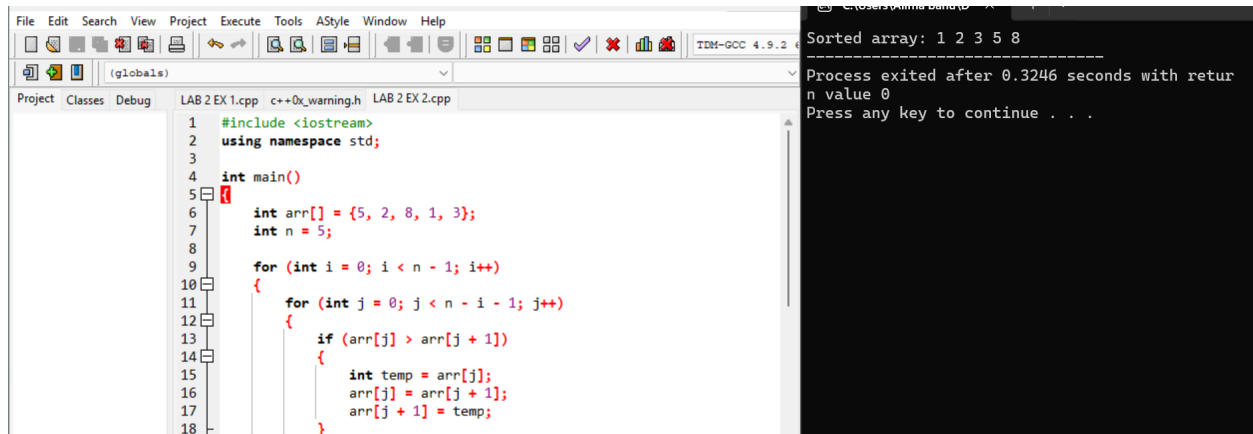
```
for (int i = 0; i < n - 1; i++)
{
    for (int j = 0; j < n - i - 1; j++)
    {
        if (arr[j] > arr[j + 1])
        {
            int temp = arr[j];
            arr[j] = arr[j + 1];
            arr[j + 1] = temp;
        }
    }
}

cout << "Sorted array: ";

for (int i = 0; i < n; i++)
{
    cout << arr[i] << " ";
}

return 0;
}
```

OUTPUT :



The screenshot shows a C++ IDE with a project named 'LAB 2 EX 1.cpp'. The code implements a bubble sort algorithm. The array {5, 2, 8, 1, 3} is sorted to {1, 2, 3, 5, 8}. The output window shows the sorted array and a message indicating the process exited after 0.3246 seconds.

```
1 #include <iostream>
2 using namespace std;
3
4 int main()
5 {
6     int arr[] = {5, 2, 8, 1, 3};
7     int n = 5;
8
9     for (int i = 0; i < n - 1; i++)
10     {
11         for (int j = 0; j < n - i - 1; j++)
12         {
13             if (arr[j] > arr[j + 1])
14             {
15                 int temp = arr[j];
16                 arr[j] = arr[j + 1];
17                 arr[j + 1] = temp;
18             }
19         }
20     }
21 }
```

Sorted array: 1 2 3 5 8

Process exited after 0.3246 seconds with return value 0

Press any key to continue . . .

3. Binary Search

```
#include <iostream>
using namespace std;

int main()
{
    int arr[] = {3, 4, 6, 8, 9, 10, 30};
    int n = 7;

    int x = 10;

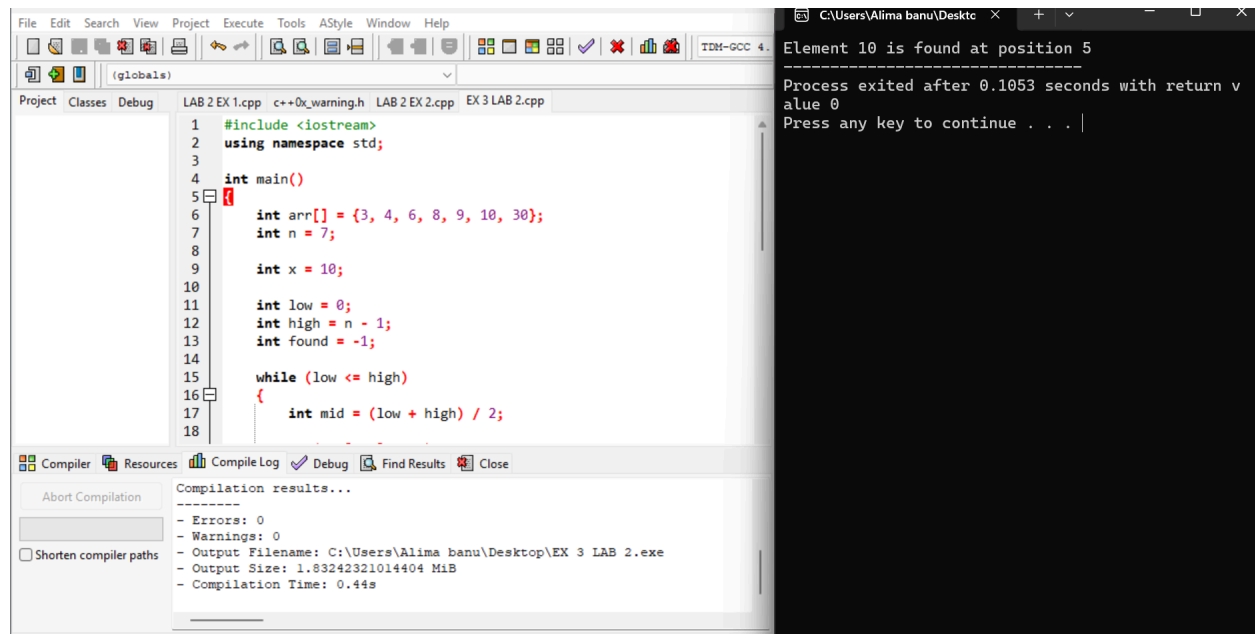
    int low = 0;
    int high = n - 1;
    int found = -1;

    while (low <= high)
    {
        int mid = (low + high) / 2;

        if (arr[mid] == x)
        {
            found = mid;
            break;
        }
    }
}
```

```
    }  
    else if (arr[mid] < x)  
    {  
        low = mid + 1;  
    }  
    else  
    {  
        high = mid - 1;  
    }  
}  
  
if (found != -1)  
{  
    cout << "Element " << x  
        << " is found at position " << found;  
}  
else  
{  
    cout << "Element " << x << " is not found";  
}  
  
return 0;  
}
```

OUTPUT :



4. Sort in $O(n \log n)$ Without Built-in Function

```
#include <iostream>
using namespace std;
```

```
void heapify(int arr[], int n, int i)
```

```
{
    int largest = i;
    int left = 2 * i + 1;
    int right = 2 * i + 2;
```

```
    if (left < n && arr[left] > arr[largest])
    {
        largest = left;
    }
```

```
    if (right < n && arr[right] > arr[largest])
```

```

    {
        largest = right;
    }

    if (largest != i)
    {
        int temp = arr[i];
        arr[i] = arr[largest];
        arr[largest] = temp;

        heapify(arr, n, largest);
    }
}

void heapSort(int arr[], int n)
{
    // Build Max Heap
    for (int i = n / 2 - 1; i >= 0; i--)
    {
        heapify(arr, n, i);
    }

    // Sort the array
    for (int i = n - 1; i > 0; i--)
    {
        int temp = arr[0];
        arr[0] = arr[i];
        arr[i] = temp;

        heapify(arr, i, 0);
    }
}

int main()
{
    int arr[] = {5, 2, 8, 1, 3, 7};
    int n = 6;

    heapSort(arr, n);

    cout << "Sorted array: ";

    for (int i = 0; i < n; i++)
    {

```

```

        cout << arr[i] << " ";
    }

    return 0;
}

```

OUTPUT :

